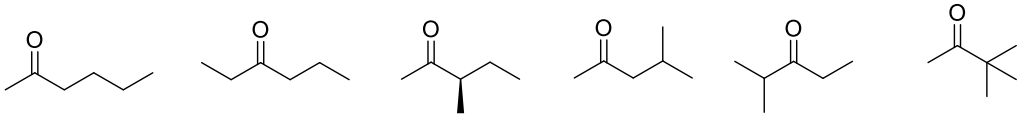
1)

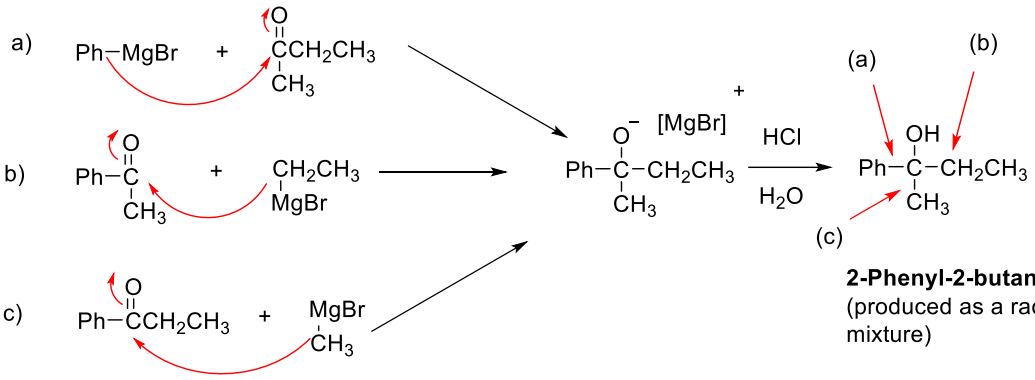
Following are the IUPAC names for the six ketones with the molecular formula $C_6H_{12}O$. Only 3-methyl-2-pentanone is chiral; the **R** enantiomer is drawn here



2- Hexanone **3- Hexanone** **(R)-3-Methyl-2-pentanone** **4-Methyl-2-pentanone** **2-Methyl-3-pentanone** **3,3-Dimethyl-2-butanone**

2)

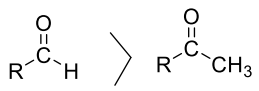
In each solution, curved arrows show formation of the new carbon-carbon bond and the alkoxide ion. The new carbon-carbon bond formed by each set of reagents is labelled in the final product.



2-Phenyl-2-butanol (produced as a racemic mixture)

3)

Alkyl groups (like CH_3) are electron donors. So based purely on electronic factors, addition to **Aldehyde** would be faster since the carbon is not as electron rich.



Aldehyde **Ketone**

Alkyl group is electron donor

C=O is more electron rich and less electrophilic

4)

Reactivity order D< C < B < A

5)

The reactivity of carbonyls depends on the presence of positive charge on carbonyl carbon and increase in sterics due to the added substituents. In this case, reactivity would decrease with increasing inductive effect neighbouring substitutes. Hence, Formaldehyde > Acetaldehyde > Acetone > Ethyl Acetate. (Check class notes)

6)

Follow notes.

7)

a) On the basis of steric crowding, the first one is most reactive, then the last one, then the middle.

b) On the basis of electronics, the middle one is most reactive, then the last, and then the first. The fluorine atom is very electronegative and pulls electron density towards itself. That leaves more positive charge on the nearby carbonyl carbon. The more fluorines on that nearby carbon, the more positive the carbon. The more positive the carbon, the more it attracts electrons from a nucleophile.